

SEMARANG MARITIME, Indonesia

WMO: 968370

Lat: 6.9508S

Lon: 110.4183E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	74.4	75.3	60.7	79.4	85.9	63.0	86.1	84.4	23.5	81.0	22.0	81.2	3.7	110	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	12.7	93.6	76.0	92.5	76.5	91.6	76.8	81.2	88.2	80.8	88.0	80.3	87.6	8.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	151.5	86.2	78.6	148.7	85.8	78.2	146.5	85.4	45.0	88.7	44.4	88.3	44.0	88.0	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	14.9	12.8	DB	71.4	96.7	2.0	2.6	70.0	98.5	68.9	100.0	67.8	101.4	66.4	103.3	(4)
(5)			WB	65.8	84.7	2.3	2.6	64.1	86.5	62.7	88.0	61.4	89.5	59.7	91.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	81.7	81.6	82.8	84.2	84.6	83.7	83.2	83.6	84.6	85.2	84.1	82.8	
	DBStd	1.92	1.69	1.68	1.41	1.45	1.52	1.43	1.37	1.32	1.55	1.88	2.04	1.66	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	12235	984	886	1018	1025	1071	1012	1028	1040	1038	1093	1022	1018	
	CDD65	6760	519	466	553	575	606	562	563	575	588	628	572	553	
	CDH74	78769	5464	4883	6302	6896	7504	6772	6538	6696	6995	7770	6759	6190	
CDH80	30502	1548	1350	2137	2745	3228	2803	2613	2750	3017	3494	2727	2090		
Wind	WSAvg	4.9	6.3	6.0	4.7	4.0	4.6	4.6	5.0	5.5	5.1	4.6	4.0	4.5	
Precipitation	PrecAvg	85.1	15.1	12.4	8.9	7.0	6.9	3.3	3.1	1.9	3.4	5.7	8.0	9.8	
	PrecMax	138.1	35.2	21.9	19.7	15.7	14.9	10.1	13.0	7.5	11.3	15.6	16.7	21.1	
	PrecMin	58.8	3.1	4.2	1.1	2.0	0.6	0.0	0.0	0.0	0.0	0.0	2.2	3.3	
	PrecStd	24.1	8.5	5.5	4.2	3.3	4.1	3.0	3.1	2.2	3.5	4.6	3.3	4.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.3	89.9	90.1	92.2	92.7	92.6	92.3	92.7	95.0	95.8	94.9	91.7	
		MCWB	79.8	79.8	79.7	79.2	78.3	76.9	75.1	74.1	73.8	75.3	77.7	79.4	
	2%	DB	88.3	88.1	88.8	91.0	91.7	91.4	91.0	91.7	93.4	93.6	92.5	89.7	
		MCWB	79.4	79.4	79.6	79.3	78.3	76.9	74.8	74.5	74.1	76.4	78.3	79.6	
	5%	DB	86.9	86.9	88.0	89.8	90.8	90.3	90.2	90.7	91.8	92.0	90.7	88.3	
		MCWB	79.0	79.1	79.4	79.4	78.4	77.0	75.2	74.9	75.0	77.2	78.9	79.3	
	10%	DB	85.9	85.8	87.1	88.7	89.7	89.3	89.1	89.5	90.2	90.5	89.0	87.2	
		MCWB	78.6	78.7	79.1	79.4	78.4	77.2	75.6	75.4	75.8	77.6	79.0	79.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	80.8	81.3	81.9	81.8	80.9	80.1	80.8	80.5	80.8	81.6	81.0
			MCDB	87.4	87.5	87.6	88.1	89.1	87.9	87.1	88.2	88.2	88.8	88.8	87.6
2%		WB	80.1	80.0	80.6	81.0	80.9	80.1	79.2	79.2	79.4	80.0	80.6	80.3	
		MCDB	86.7	86.4	87.2	87.8	88.4	87.5	86.6	86.8	87.3	88.3	88.7	87.4	
5%		WB	79.4	79.4	80.0	80.5	80.2	79.6	78.5	78.3	78.7	79.5	79.9	79.8	
		MCDB	85.6	85.5	86.5	87.5	87.8	87.0	86.1	86.0	86.8	87.9	87.9	86.7	
10%		WB	78.9	79.0	79.5	80.0	79.7	79.0	77.9	77.7	78.0	78.9	79.4	79.2	
		MCDB	84.9	84.8	85.8	86.8	87.1	86.5	85.3	85.5	86.1	87.5	87.2	85.9	
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.1	8.7	10.1	11.2	12.0	13.0	13.9	14.1	12.7	11.1	9.1	
		MCDBR	9.7	9.5	9.8	11.3	12.1	13.5	14.2	14.7	15.8	14.5	13.3	10.6	
		MCWBR	4.3	4.3	4.4	4.4	4.5	5.4	6.1	6.7	6.7	5.5	5.2	4.6	
	5% WB	MCDBR	9.1	8.8	9.3	10.2	10.9	11.2	11.9	13.0	13.5	12.6	11.6	9.7	
		MCWBR	4.3	4.3	4.4	4.4	4.5	4.6	4.9	5.5	6.6	6.1	5.6	5.2	4.5
Clear-Sky Solar Irradiance	taub	0.533	0.539	0.539	0.549	0.522	0.511	0.514	0.545	0.576	0.656	0.639	0.575		
	taud	2.099	2.084	2.085	2.041	2.107	2.128	2.106	2.002	1.925	1.741	1.792	1.963		
	Ebn at Noon	260	258	254	243	240	238	240	240	241	227	231	247		
	Edh at Noon	54	55	54	54	49	46	48	56	62	77	73	62		
All-Sky Solar Radiation	RadAvg	1449	1467	1592	1629	1605	1543	1643	1835	1965	1873	1656	1459		
	RadStd	146	128	97	98	89	103	99	69	109	166	156	124		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.58	N/A	N/A	+0.70	+0.63	+0.85	N/A	N/A	N/A	N/A

Nomenclature: See separate page